GLOBAL ECONOMICS
Grade: 11TH
Learning Evidence 1
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3rd TERM
2026–2027

Learning objective:	**Assessment criteria**
Characterise continuous probability distributions and apply appropriate models to calculate and interpret probabilities in economic and financial contexts.	C1. Characterises continuous random variables by distinguishing them from discrete variables and interpreting their probability density function, cumulative distribution function, and support in economic contexts. C2. Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas. C3. Applies circular-arc probability density functions to calculate interval probabilities using the geometry of circles and circular segments. C4. Applies the exponential distribution to constant-rate waiting-time situations and assesses its suitability.

Criteria assessment

This task sheet assesses criteria C1 through C4. Each problem assesses the criteria indicated in its title. The number of points assigned to each task is shown at the end of the item. Partial credit may be awarded when the correct procedure is shown but arithmetic errors are made. Answers that do not show the required procedure will not receive credit. A minimum of 4 points is required to pass each criterion.

Question	**C1**	**C2**	**C3**	**C4**
1	5	–	–	–
2	–	3	–	–
3	–	4	–	–
4	–	–	2	–
5	–	–	3	–
6	–	–	–	0.5
7	–	–	–	0.5
8	–	–	–	1
9	–	–	–	1
10	–	–	–	1
11	–	–	–	2
Total	**5**	**7**	**5**	**6**

1. Identifying Probability Distributions from Contexts [C1-5]

A variety of Colombian organizations use probability models to represent uncertain quantities and events. In each situation below, a random variable is defined from the information provided. The appropriate probability distribution must be identified from the characteristics of the situation. For each context, characterize the random variable and identify the appropriate probability distribution.

Context 1. A university sends an invitation to exactly 60 randomly selected alumni to participate in a fundraising campaign. Each alumnus either makes a donation or does not. The university records the number of alumni who donate to estimate the expected response to the campaign.

Context 2. A courier company models the travel time for a delivery route. Travel times range from 25 to 55 minutes, with times around 40 minutes occurring most frequently. Values become progressively less common as they move away from 40 minutes toward either endpoint. The company uses this model to estimate delivery durations.

Context 3. A telecommunications company counts the number of incoming technical-support requests received by a service center during a fixed 30-minute period. Historical records indicate that requests occur independently and without a predictable pattern. The company uses this model to plan the number of available support agents.

Context 4. A water-utility company studies the daily water consumption of a small residential building. Consumption ranges from 100 to 500 litres, and the probability density increases steadily as consumption approaches 500 litres. The company uses this model to plan daily supply capacity.

Context 5. A manufacturer examines batches of circuit boards until the first board that fails a functional test is found. Boards are tested independently, and testing stops immediately after the first failure. The manufacturer records the number of boards tested, including the failed board, to estimate inspection requirements.

Context 6. A logistics company studies the delay, in minutes, of trucks arriving at a distribution center. Delays can take any value between 0 and 30 minutes, with all values in this interval considered equally likely. The company uses this model to estimate scheduling margins.

Context 7. A hospital records the number of patients arriving at an emergency department during each fixed four-hour shift. Comparable shifts show that arrivals occur independently and without a predictable timetable. The hospital uses this model to plan staffing levels.

Context 8. A cinema chain sends a discount code to exactly 150 randomly selected customers. Each customer either uses the code or does not. The chain records the number of customers who use the discount to estimate the additional attendance generated by the campaign.

Context 9. A dairy producer measures the fat content of milk samples from its regular production process. Measurements cluster around 3.5%. Measurements farther from 3.5% occur progressively less often. The producer uses this model to establish quality-control limits.

Context 10. A city's electric utility studies the daily electricity consumption of a small commercial building. Consumption ranges from 200 to 800 kWh, and the probability density decreases steadily as consumption moves from 200 kWh toward 800 kWh. The utility uses this model to estimate the likelihood of different daily loads.

2. Delivery Time for a Local Grocery Order [C2-3]

A transportation company studies the additional charging time required for electric vehicles at one of its charging stations. For a particular type of vehicle, the additional charging time can range from 0 to 4 minutes. A charging time of any length within this interval is possible, and shorter additional times become steadily more common. Historical data indicate that an additional charging time of 0 minutes is twice as common as an additional charging time of 4 minutes. Determine the probability that the additional charging time is between 1 and 3 minutes.

3. Daily Commute Time for a Bogotá Apprentice [C2-4]

A local business records its daily sales revenue over several months. Daily revenue ranges from \$1 million to \$7 million. Revenue becomes steadily more common from \$1 million up to a most common level of \$6 million, then steadily less common through \$7 million; there are no observations exactly at either limit. Determine the probability that the daily sales revenue is between \$2 million and \$6 million.

4. Timing of an Unexpected Market Movement [C3-2]

A retail company monitors the time of day at which an unexpected price adjustment occurs in one of its online markets. The adjustment is equally likely to occur at any time during a 24-hour period, so no part of the day is more likely than another. Determine the probability that the price adjustment occurs between 8 p.m. and 6 a.m.

5. Direction of an Unexpected Sensor Reading [C3-3]

A directional sensor records the direction in which it is pointing. The sensor can point in any direction around a complete circle, and every direction is equally likely. Let A represent the direction of the sensor, measured in radians from a fixed reference marker at 0 radians. Determine which is more probable and by how much: the sensor pointing within $\pi/10$ radians of the reference marker, or the sensor pointing within 15° of the reference marker.

6. Waiting for a Study-Room Opening [C4-0.5]

A food-delivery platform receives driver arrivals at an average rate of 5 drivers every 8 minutes. Determine the probability that the next driver arrives more than 2 minutes after the start of the observation.

7. Waiting for a Market Price Alert [C4-0.5]

A public-transport monitoring system detects significant service delays at an average rate of 4 alerts every 15 minutes. Determine the probability that the next significant delay is detected within 3 minutes after the start of the observation.

8. Waiting for a Demand Alert [C4-1]

A supply chain monitoring system detects significant disruptions at an average rate of 5 alerts every 20 minutes. Determine the probability that the next significant disruption is detected between 4 and 8 minutes after the observation time.

9. Processing Investment Transactions [C4-1]

An online store receives customer orders throughout the day. The time between consecutive orders is modeled by an exponential distribution. Based on previous observations, there is a 60% probability that the next order is received within 5 minutes. On average, how many orders are expected to be received in a 12-minute interval?

10. Waiting for a Loan Approval [C4-1]

An economic analysis system monitors market indicators throughout the day. The time between consecutive significant indicator updates is modeled by an exponential distribution. Based on previous observations, there is a 25% probability that the next significant update takes more than 10 minutes to occur. Find the average number of significant updates expected per minute.

11. Comparing Investment Transaction Rates [C4-2]

wo economic forecasting systems monitor significant changes in market indicators throughout the day. The detected changes by each system are modeled by independent Poisson processes.

For System A, there is a 70% probability that the next significant change is detected within 10 minutes. For System B, there is a 50% probability that the next significant change is detected within 6 minutes. Determine after how many minutes the expected numbers of significant changes detected by the two systems differ by 1 change.